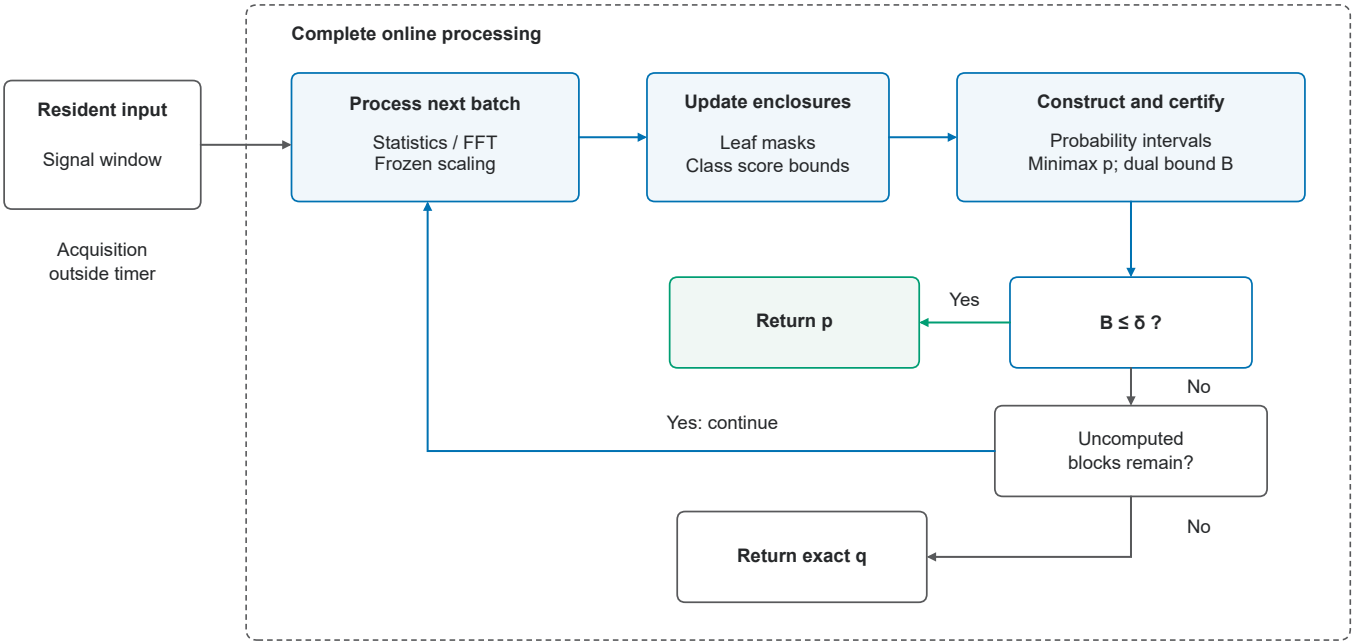
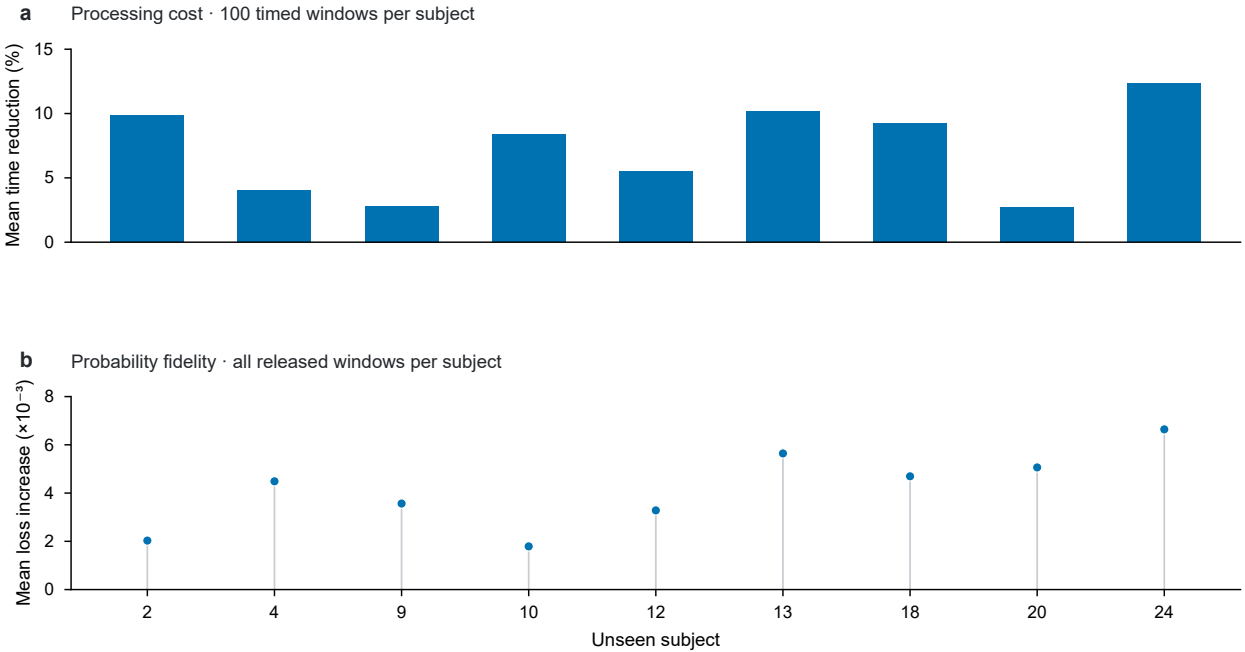
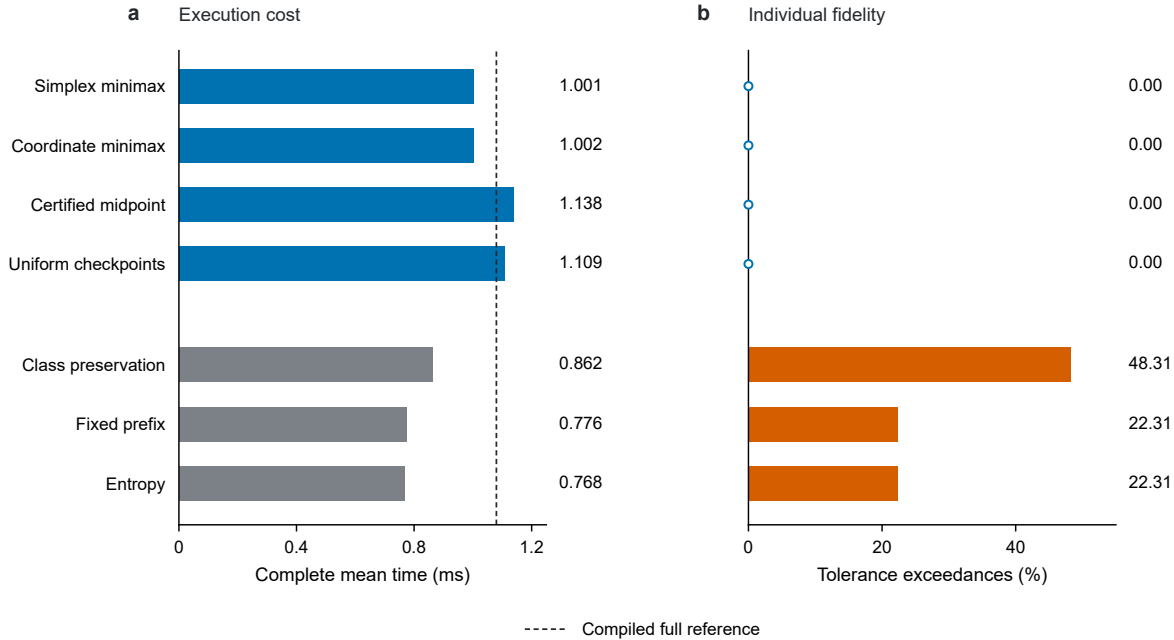
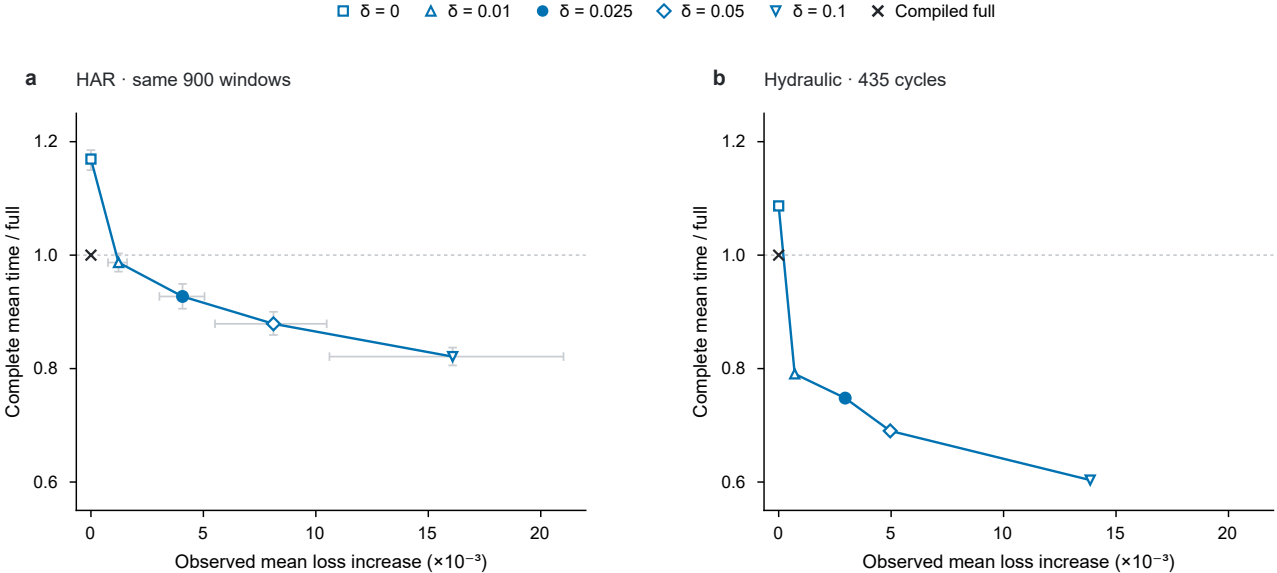
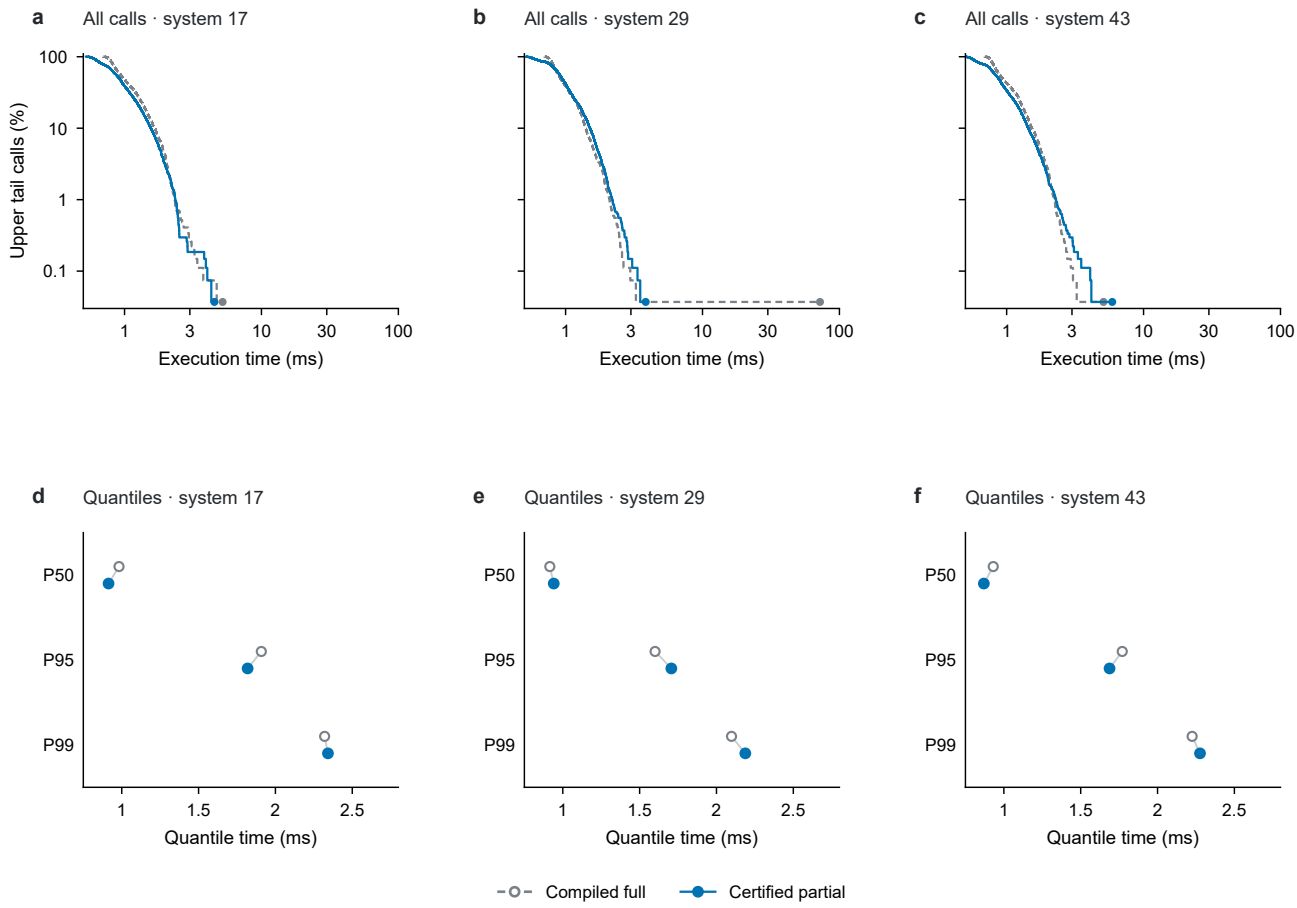

Complete online processing
Resident input
Signal window
Acquisition
outside timer
Process next batch
Statistics / FFT
Frozen scaling
Update enclosures
Leaf masks
Class score bounds
Construct and certify
Probability intervals
Minimax p; dual bound B
B ≤ δ ?
Yes
Return p
No
Yes: continue
Uncomputed
blocks remain?
No
Return exact q
Timer: state + signal operators + enclosures + certificate + control + exact fallback


a
Processing cost · 100 timed windows per subject
Mean time reduction (%)
15
10
5
0
b
Probability fidelity · all released windows per subject
Mean loss increase (×10⁻³)
8
6
4
2
0
2
4
9
10
12
13
18
20
24
Unseen subject


a
Execution cost
Simplex minimax
Coordinate minimax
Certified midpoint
Uniform checkpoints
Class preservation
Fixed prefix
Entropy
1.001
1.002
1.138
1.109
0.862
0.776
0.768
0
0.4
0.8
1.2
Complete mean time (ms)
b
Individual fidelity
0.00
0.00
0.00
0.00
48.31
22.31
22.31
0
20
40
Tolerance exceedances (%)
Compiled full reference


δ = 0
δ = 0.01
δ = 0.025
δ = 0.05
δ = 0.1
Compiled full
a
HAR · same 900 windows
b
Hydraulic · 435 cycles
Complete mean time / full
1.2
1.0
0.8
0.6
0
5
10
15
20
Observed mean loss increase (×10⁻³)


a
All calls · system 17
b
All calls · system 29
c
All calls · system 43
Upper tail calls (%)
100
10
1
0.1
1
3
10
30
100
Execution time (ms)
d
Quantiles · system 17
e
Quantiles · system 29
f
Quantiles · system 43
P50
P95
P99
1
1.5
2
2.5
Quantile time (ms)
Compiled full
Certified partial

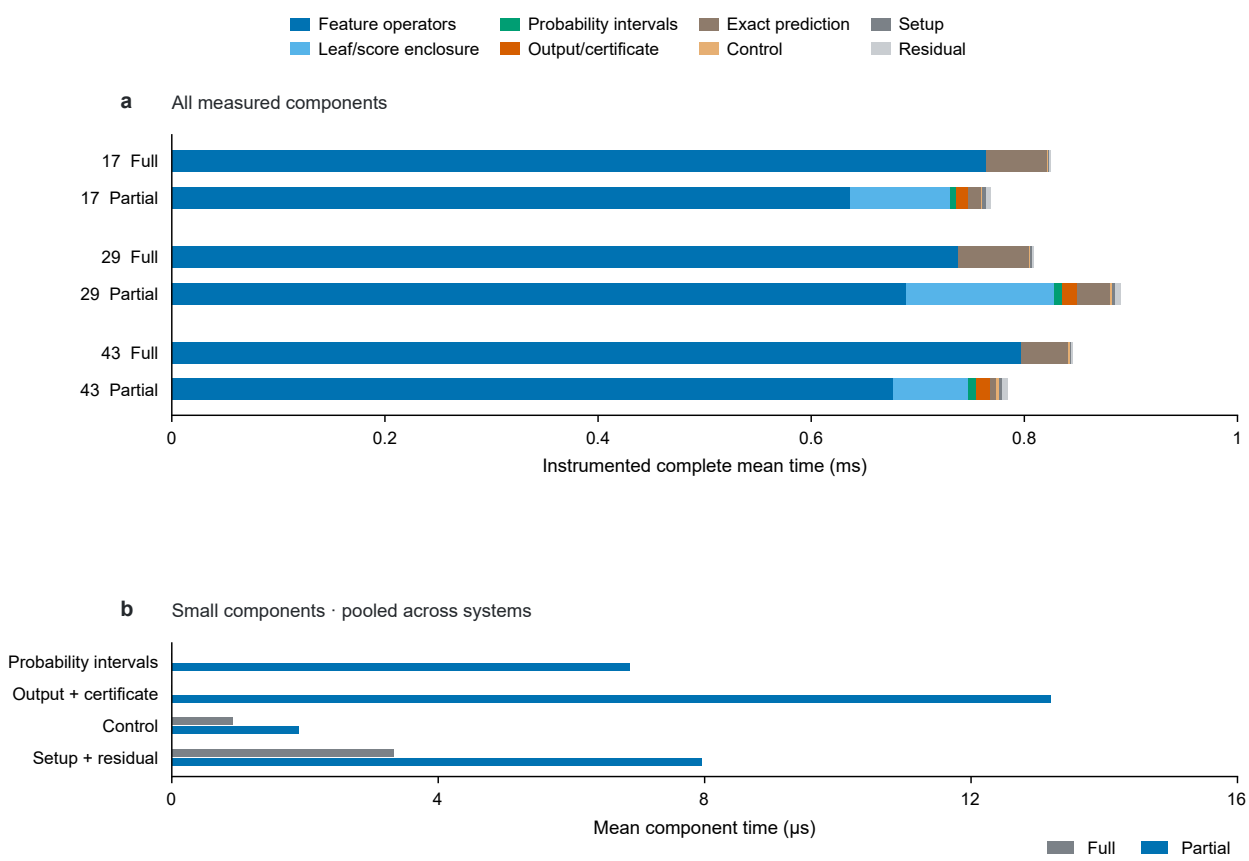

Feature operators
Leaf/score enclosure
Probability intervals
Output/certificate
Exact prediction
Control
Setup
Residual
a
All measured components
17 Full
17 Partial
29 Full
29 Partial
43 Full
43 Partial
0
0.2
0.4
0.6
0.8
1
Instrumented complete mean time (ms)
b
Small components · pooled across systems
Probability intervals
Output + certificate
Control
Setup + residual
0
4
8
12
16
Mean component time (µs)
Full
Partial

Table 1 | Related capabilities and the guarantee being evaluated

Research line	Main object addressed	Relation to the present method
Active feature acquisition [12–16]	Predictive utility and observation cost under subset observations	May supply a feature schedule; does not substitute for checking the stated relative-loss certificate
Risk-controlled early exit [5]	Calibrated distributional risk or a final-model performance gap	Establishes the relative Brier target; the quantifier here is individual input and every label
Certified lazy boosted-tree inference [6]	Preservation of the complete winning class	Supplies a distinct output criterion, tested by score separation
Adversarial uncertainty certification [7]	Worst calibration quantities over a perturbation set	Related certification tools with a different domain and loss object
Present construction	Worst relative multiclass half-Brier loss over complete outputs consistent with observed blocks	Constructs and certifies a probability vector, while charging its full processing cost

The present objective is pointwise relative multiclass half-Brier fidelity to the complete predictor.

Table 2 | Signal-processing workloads and frozen evaluation cohorts

Source	Task	Computed representation	Fitting/development	Frozen evaluation
Hydraulic	Four valve-condition classes	14 channels; 28 blocks; 238 features	907 / 223 cycles	435 previously held-out cycles, grouped by operating-segment proxies
Smartphone HAR	Six activity classes	9 channels; 18 blocks; 153 features	17 / 4 official training subjects per system	2,947 windows from nine official test subjects

HAR uses the official subject split. Hydraulic grouping uses operating-segment proxies, not documented independent runs.

Table 3 | Complete-model quality on the held-out HAR release

Complete family	System 17: half-Brier / accuracy	System 29: half-Brier / accuracy	System 43: half-Brier / accuracy	Mean half-Brier / accuracy
HGB, certified method's reference	0.056736 / 91.79%	0.077797 / 90.33%	0.049557 / 93.59%	0.061363 / 91.90%
Random forest	0.071221 / 90.74%	0.079527 / 88.97%	0.060011 / 92.37%	0.070253 / 90.69%
Extra trees	0.068181 / 91.21%	0.076067 / 90.50%	0.064143 / 92.09%	0.069464 / 91.27%
Logistic regression	0.069117 / 90.67%	0.094257 / 88.60%	0.056360 / 92.60%	0.073245 / 90.62%
RBF SVC	0.050195 / 93.28%	0.063627 / 91.86%	0.042154 / 94.67%	0.051992 / 93.27%

All 2,947 held-out windows; each system is a separately frozen training-subject realization. Accuracy is a percentage.

Table 4 | Complete-model processing cost and same-cohort quality

Complete implementation	Mean / median time, ms	P95 / P99, ms	Same-cohort half-Brier	Same-cohort accuracy
Native HGB	5.935996 / 5.636200	9.005495 / 10.461700	0.066411	91.11%
Compiled HGB	1.149548 / 1.049300	1.802545 / 2.151405	0.066411	91.11%
Random forest	14.798855 / 13.466150	22.953085 / 26.324272	0.075713	89.85%
Extra trees	15.155591 / 13.812650	23.348470 / 26.783967	0.073094	90.74%
Logistic regression	1.295108 / 1.196150	2.024415 / 2.360212	0.079415	89.67%
RBF SVC	1.625706 / 1.498900	2.519920 / 2.975855	0.056553	92.56%

Same 900 windows, three systems and three randomized repetitions per implementation. Complete time includes all 18 raw-signal blocks; this is a separate benchmark from the primary partial/full experiment.

Table 5 | Primary HAR comparison at half-Brier tolerance 0.025

System	Complete full time, ms	Partial time, ms	Time reduction	Partial / full half-Brier	Mean blocks of 18	Full fallback
17	1.122237	1.016754	9.40%	0.066525 / 0.063736	14.5089	19.89%
29	1.051749	1.023604	2.68%	0.090309 / 0.083996	15.7778	38.22%
43	1.066047	0.963610	9.61%	0.054615 / 0.051502	14.6711	17.33%
Mean	**1.080011**	**1.001322**	**7.29%**	**0.070483 / 0.066411**	**14.9859**	**25.15%**

Same 900 windows per system. Complete time includes feature processing, enclosure, certification, control and exact fallback. Mean is the equal-system pooled summary; blocks and fallback refer to partial processing.

Table 6 | HAR stopping and output-construction controls

Procedure	Complete mean time, ms	Time reduction	Half-Brier	Observed-label mean increase	Every-label exceedances / 2,700
Compiled full	1.080011	—	0.066411	0	0
Simplex-aware minimax, 0.025	1.001322	7.29%	0.070483	0.004072	0
Coordinate minimax, 0.025	1.001969	7.23%	0.070369	0.003958	0
Midpoint with tight certificate, 0.025	1.137513	−5.32%	0.066579	0.000168	0
Minimax with uniform checkpoints, 0.025	1.108992	−2.68%	0.072028	0.005617	0
Score-separated class preservation	0.862145	20.17%	0.093246	0.026834	1,289
Fixed-prefix mean completion	0.775521	28.19%	0.073406	0.006995	641
Entropy mean completion	0.768284	28.86%	0.073406	0.006995	641

Same 900 windows for each of three systems; 2,700 system–windows in the exceedance denominator. Mean observed-label increase and every-label exceedance are distinct criteria.

Table 7 | Prescribed fidelity tolerance and measured complete time

	Smartphone HAR			Hydraulic confirmation		
Half-Brier tolerance	Complete time, ms	Reduction	Same-cohort mean increase	Complete time, ms	Reduction	Mean increase
0	1.262918	−16.94%	0	3.108712	−8.68%	0
0.01	1.065939	1.30%	0.001226	2.261661	20.93%	0.000713
0.025	1.001322	7.29%	0.004072	2.139557	25.20%	0.002957
0.05	0.949258	12.11%	0.008115	1.974194	30.98%	0.004959
0.10	0.886820	17.89%	0.016082	1.726200	39.65%	0.013851

HAR uses the same 900-window timing cohort across all three systems. Hydraulic uses all 435 confirmation cycles. Reductions are against contemporaneous compiled full execution.

Table 8 | Complete and partial latency quantiles

Evaluation system	Full / partial P50, ms	Full / partial P95, ms	Full / partial P99, ms
HAR 17	0.982250 / 0.914200	1.908330 / 1.819120	2.320208 / 2.341802
HAR 29	0.915500 / 0.941050	1.600410 / 1.706305	2.097817 / 2.187748
HAR 43	0.930950 / 0.869850	1.770430 / 1.687745	2.225132 / 2.276310
Hydraulic confirmation	2.509400 / 1.943500	4.245220 / 3.986300	4.843356 / 4.813808

Each pair is full / partial. HAR quantiles use each system's 2,700 recorded calls; Hydraulic uses 1,305 calls. Quantiles describe measured latency, not hard real-time bounds.

Table 9 | Primary HAR latency conditioned on processing outcome

Processing outcome	System–windows	Partial / full mean, ms	Partial-minus-full mean, ms	Partial slower
Stop at 12 blocks	1,024	0.781742 / 1.077781	−0.296039	4.10%
Stop at 16 blocks	997	1.054817 / 1.072119	−0.017302	45.34%
18 blocks, exact fallback	679	1.253923 / 1.094962	+0.158962	84.09%

2,700 system–window pairs; each mean first averages three repetitions for the same system–window. Partial slower is the fraction whose paired mean is larger, not a causal effect of fallback.

Table 10 | Hydraulic confirmation at half-Brier tolerance 0.025

Hydraulic procedure at tolerance 0.025	Complete time, ms	Reduction against compiled full	Half-Brier	Mean blocks	Full fallback
Compiled full	2.860459	—	0.126473	28.0000	—
Simplex-aware minimax	2.139557	25.20%	0.129430	19.0069	19.77%
Coordinate minimax	2.070535	27.62%	0.129353	19.0345	19.77%
Midpoint with tight certificate	2.642566	7.62%	0.126445	23.4299	35.40%

435 frozen confirmation cycles; complete time includes all online work. Mean blocks are out of 28. The coordinate certificate is cheaper here despite its looser relaxation.

Table 11 | Development-only sensitivity to feature-block order

Order	Mean blocks of 18	Exact fallback	First-12-block sum of interval widths	First-12-block diameter upper proxy
Locked gain/time	14.7179	18.33%	0.623226	0.340698
Descending gain only	14.5794	14.59%	0.594592	0.326628
Reverse locked order	18.0000	100.00%	5.670742	1.414214
One fixed random order per system	17.1791	68.84%	3.468061	1.246648

Equal-system means on the original development subjects; no new raw-signal timing or test-policy selection. Diameter upper proxy is computed per case before averaging and does not measure the exact enclosure gap.

Table 12 | Requirements for extending the partial enclosure

Complete predictor	Route to a sound partial enclosure	Additional requirement
Random forest / extra trees	Bound each leaf's class-probability vector and average tree bounds	Preserve probability averaging and the frozen temperature transformation; a boosted-logit sum is not equivalent
XGBoost / LightGBM multiclass trees	Sum reachable-leaf bounds for each class score, then bound the link	Validate intercepts, class/tree indexing, missing and categorical splits, and libraries' exact output semantics
Larger numerical trees	Use multiple mask words or another candidate-leaf representation	Charge extra update and interval-extraction work; the one-word runtime does not transfer automatically
RBF SVC	Enclose kernels and decision values under partial features	Also enclose the fitted pairwise probabilities and multiclass coupling; the present softmax score formula cannot simply replace native SVC semantics
Neural network	Propagate sound feature bounds to logits, then probabilities	Justify bounds for uncomputed signal features and control relaxation cost; empirical training minima/maxima alone do not enclose unseen inputs

These routes are proposed extensions. Only numerical multiclass HGB trees with at most 63 leaves per tree were implemented and evaluated.